Name [redacted] **ID#** ______________________

Math 150 Spring 2025 Final Exam

- Desmos is allowed
- Desmos and the only app (or website) allowed - no other apps or websites are allowed at anytime, this is **not** an open internet test.
- Must show all work to receive credit
- Clearly label all graphs and make sure both axes have a clear scale.
- Attach all scratch paper

	Question	Points
1	Integration	5 /5
2	Riemann Sums	4 /5
3	Volume	5 /5
4	Limits & Continuity	3 /~~4~~ 5
5	Average Value	5 / ~~4~~ 5
6	Optimization	4 /~~4~~ 5
7	Position, Velocity, Acceleration	4 /~~4~~ 5
8	Related Rates	~~/3~~
9	First Derivative Test	~~/3~~
10	Geometry	5 /~~3~~ 5
Total		35 / 40

~~5~~ ·interval of interest

1) Evaluate the integral $\int 4\sin^5(x)\cos^3(x)\,dx$ using two different substitutions and discuss how the two anti-derivatives are related.

$4\int \sin^5(x)\cos^3(x)$ $\quad \sin = u \quad \cos\, du$

$\cos(\sin^5(x))(\cos^2)$

$\cos(\sin^5 x)(1-\sin^2 x)$

$du\,(u^5)(1-u^2)$

$4\int (u^5 - u^7)\,du$

$4\left|\frac{u^6}{6} - \frac{u^8}{8}\right.$

$\frac{2\sin^6 x}{3} - \frac{\sin^8 x}{2} + C$

$4\int \sin^5(x)\cos^3(x)$

$4\int \sin(1-\cos^2(x))^2(\cos^3 x)$

$4\int du\,(1-u^2)^2(u^3)$ $\quad u = \cos$, $du = \sin$

$4\int (u^4 - 2u^2 + 1)\,u^3$

$u^7 - 2u^5 + u^3$

$4\left(\frac{\cos^8 x}{8} - \frac{\cos^6 x}{3} + \frac{\cos^4 x}{4}\right) + C$

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2) Estimate the signed area under the curve $y = 4 - x^2$ from 0 to 3 using:

(a) three (3) rectangles with the **left endpoint** approximation.

(b) three (3) rectangles with the **right endpoint** approximation,

(c) three (3) rectangles with the **midpoint** approximation,

(d) Find the exact signed area using FTC and compare your estimates from parts (a-c)

In each case a-c draw the graph and rectangles and use sigma notation.

$$\int_0^2 (4-x^2) + \int_2^3 0-(4-x^2)$$

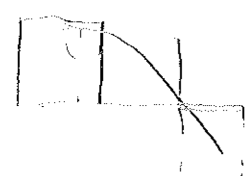
$$\left[4x - \frac{x^3}{3}\right]_0^2 + \int_2^3 -4x + x^2 = \left[4x - \frac{x^3}{3}\right]_0^2 + \left[-4x + \frac{x^3}{3}\right]_2^3$$

$$\left(4(2) - \frac{8}{3}\right) + \left[\left(-4(3) + \frac{3^3}{3}\right) - \left(-4(2) + \frac{2^3}{3}\right)\right]$$

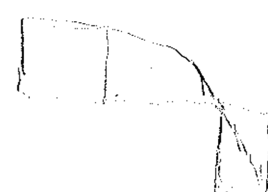
$$\frac{16}{3} + 1 = \frac{7}{3} + \frac{16}{3} = \frac{23}{3} \approx 7.667$$

0, 1, 2, 3, 4

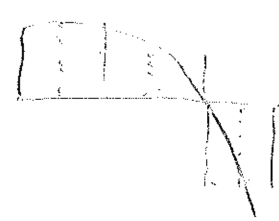
$$\sum_{n=1}^{3} \left(4 - (n-1)^2\right) = \boxed{7}$$



$$\sum_{n=1}^{3} \left(4 - (n)^2\right) = \boxed{-2}$$



$$\sum_{n=1}^{3} \left(4 - \left(n - \frac{1}{2}\right)^2\right) = \boxed{3.25}$$

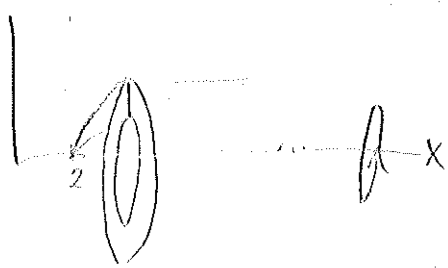


$e^{1 = \ln\frac{x}{2}} \quad 2e^1 \qquad e^{1 = \ln\frac{x^2}{4}} \quad e \quad \frac{x^2}{4} \qquad \sqrt{4e} = x$

$2\sqrt{e}$

3) Let R be the region bounded by the curves $y = \sqrt{\ln\left(\frac{x}{2}\right)}$, $y = \sqrt{\ln\left(\frac{x^2}{4}\right)}$ and $y = 1$.

a) Express the volume of R revolved around the x-axis using Washers.
b) Draw the region R, a typical slice, and the washer or disk generated by the slice revolving around the x-axis.
c) Express the volume of R revolved around the x-axis using Shells.
d) Draw the region R, a typical slice, and the shell generated by the slice revolving around the x-axis.
e) Choose one of the two integrals to calculate the exact volume.



$V = \pi \int (f(x))^2 - g(x)^2$

$V = \pi \int_2^{\sqrt{4e}} \ln\left(\frac{x^2}{4}\right) - \ln\left(\frac{x}{2}\right) + \pi \int_{\sqrt{4e}}^{2e} 1 - \ln\left(\frac{x}{2}\right)$

$y = \sqrt{\ln\left(\frac{x}{2}\right)} \quad y^2 = \ln\left(\frac{x}{2}\right) \quad \boxed{2e^{y^2} = x}$

$y = \sqrt{\ln\left(\frac{x^2}{4}\right)} \quad y^2 = \ln\left(\frac{x^2}{4}\right) \quad 4e^{y^2} = x^2 \quad \sqrt{4e^{y^2}} =$

$2\pi \int_0^1 y\left(2e^{y^2} - \sqrt{4e^{y^2}}\right)$

$y2e^{y^2} - y2\left(e^{y^2}\right)^{\frac{1}{2}}$

$u = y^2$

$du = 2y$

$u = y^2$

$du = 2ye^{y^2}$

$2\pi \int_0^1 e^u - (e^u)^{\frac{1}{2}} \, du$

$\boxed{V = 2.644}$

$2\pi \left| e^u - \frac{3(e^u)^{\frac{3}{2}}}{2} \right.$

$e^1 - 3e^1$

$2\pi \int_0^1 e^u - \frac{9(e^u)^{\frac{3}{2}}}{3}$

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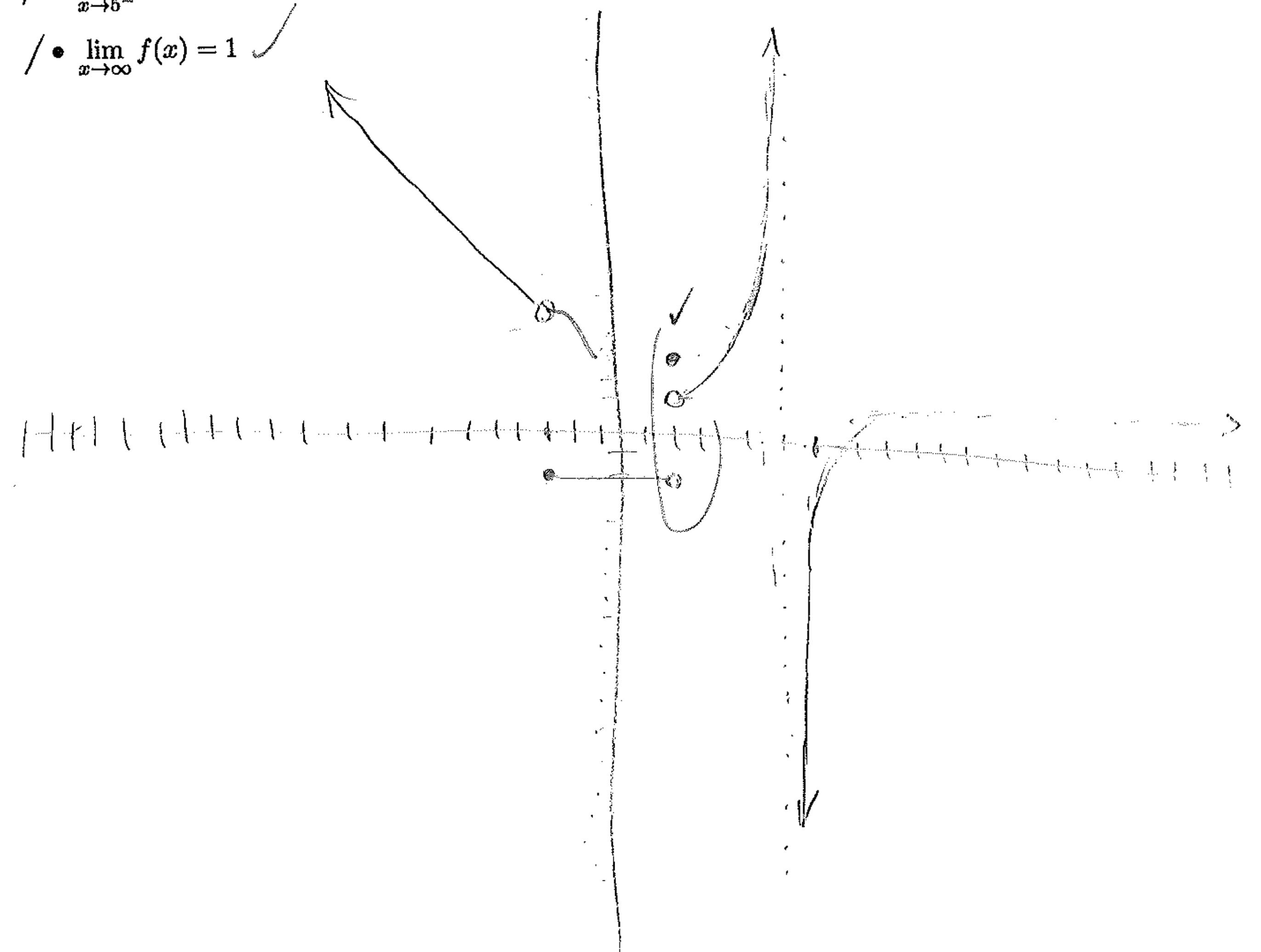
4) Sketch the graph of a function with following limits, and classify any discontinuities.

- $\lim_{x \to -3} = 5$
- $f(-3) = 2$
- $\lim_{x \to 2^-} f(x) = -2$
- $\lim_{x \to 2^+} f(x) = 1$
- $f(2) = 3$
- $\lim_{x \to 5^+} f(x) = \infty$
- $\lim_{x \to 5^-} f(x) = -\infty$
- $\lim_{x \to \infty} f(x) = 1$



5) Find the exact average value of the function $f(x) = \tanh(x)$ on the interval $-1 \leq x \leq 2$.

Graph the function and indicate the approximate average value on the graph.

$$\frac{1}{b-a}\int_a^b f(x) \qquad \frac{1}{2+1}\int_{-1}^{2} \tanh(x)$$

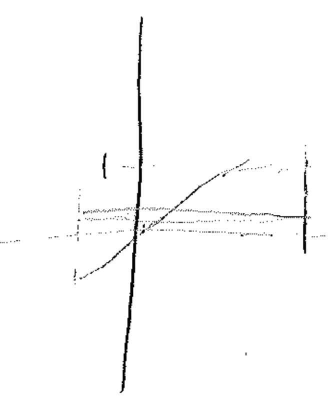
$$\frac{1}{3}\int_{-1}^{2} \frac{\sinh(x)}{\cosh(x)} \qquad \frac{1}{3}\int \frac{1}{u}\,du$$

$u = \cosh x$
$du = \sinh(x)$

$$\frac{1}{3}\Big|_{-1}^{2} \frac{1}{u}\,du \qquad \frac{1}{3}\Big|_{-1}^{2} \ln u \qquad \frac{1}{3}\Big|_{-1}^{2} \ln|\cosh x|$$

$$\frac{1}{3}\left[\ln\left(\frac{e^{2}+e^{-2}}{2}\right) - \ln\frac{e^{-1}+e^{1}}{2}\right]$$

$$\frac{1}{3}\ln\left(\frac{e^{2}+\frac{1}{e^{2}}}{e+\frac{1}{e}}\right) \approx 0.29$$



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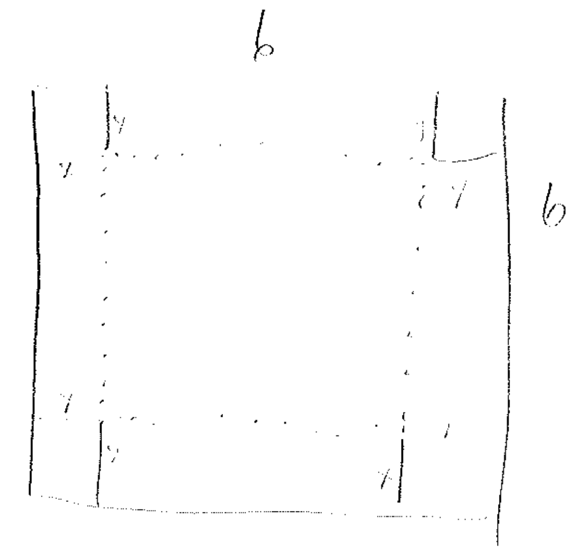
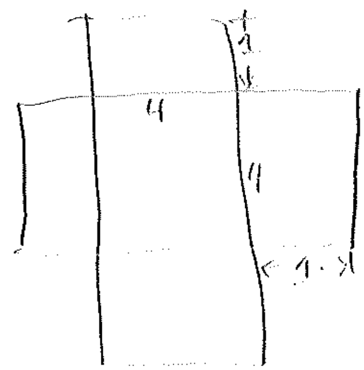
6) A sheet of paper measuring 6in by 6in is constructed into an open top box by removing four corners. Find the exact dimensions of the box with maximum volume. State the objective function and interval of interest.





$x(6-2x)(6-2x)$ $\quad [0, \cancel{\infty}]$

$x(36-24x+4x^2)$

$(36x-24x^2+4x^3)$

$36-48x+12x^2=0$

$x=1 \quad x=3$

$\frac{-b\pm\sqrt{b^2-4ac}}{2a} \quad \frac{48\pm\sqrt{-48^2-4(36)(12)}}{2(36)}$

$\cancel{3(6-6)(6-6)}$

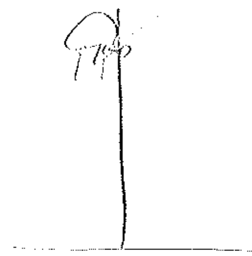
$1(6-2)(6-2)$

$\boxed{1(4)(4)=16}$

P(t)
V(t)
A(t)

$A(t) = -9.8$

7) A payload is released at an elevation of 700 meters from a hot air ballon that is rising at a rate of 11 meters per second. Assume no wind resistance and acceleration due to gravity is 9.8 meters per second squared.



a) Find the velocity of the payload. $v(t) =$

$V(t) = -9.8t + 11$

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b) Find the position of the payload. $s(t) =$

$S(t) = -4.9t^2 + 11t + 700$

c) The payloads highest point is 706.17 meters above the ground at time = ~~13.276~~ seconds.

$V(0) = -9.8t + 11$

$\left(\frac{11}{9.8}\right) = V = 0$

$S\left(\frac{11}{9.8}\right) = -4.9\left(\frac{11}{9.8}\right)^2 + 11\left(\frac{11}{9.8}\right) + 700 = 13.216$

d) Find the time when the payload strikes the ground.

$\frac{-11 \pm \sqrt{11^2 - 4(-4.9)(700)}}{2(-4.9)}$

10.889

$\boxed{13.276}$

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8) A block of ice in the shape of a perfect cube is melting. Each side of the ice cube is decreasing at a constant rate of 5 mm per minute.

a) How quickly is the surface area of the ice cube shrinking when each side is 50 mm? $\frac{1}{1500}$ mm/m

b) How quickly is the volume of the ice cube shrinking when each side is 50 mm?

5 mm/m V. 5 mm/m $\frac{V}{dt}$

$V = s^3$

$\frac{dV}{dt} = 3s^2 \frac{ds}{dt}$

$5 \text{ mm/m} = 3(50)^2 \frac{ds}{dt}$

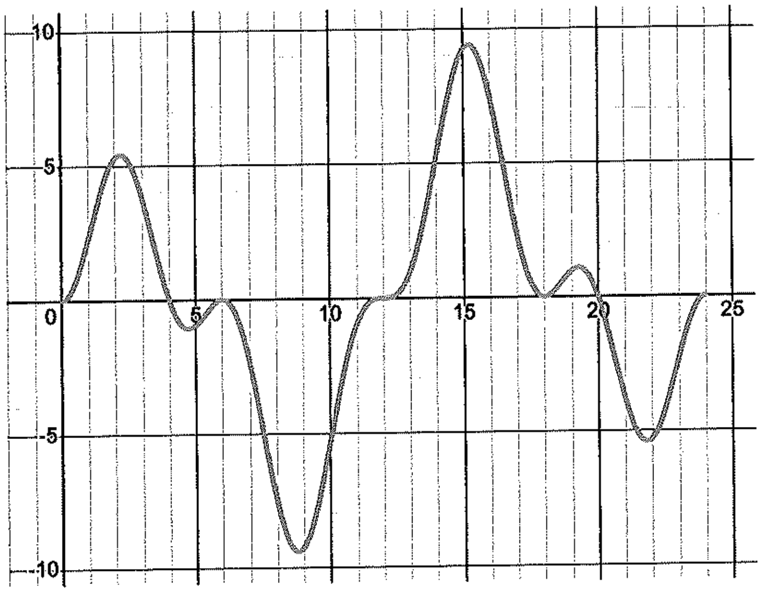
$5 \text{ mm/m} = 3(2500) \frac{ds}{dt}$

$\frac{5}{7500} = \frac{ds}{dt}$

$\frac{1}{1500} = \frac{ds}{dt}$

9) Given the graph of power $P(t)$ over a 24 hour period below, and that power $P(t) = E'(t)$.

Graph energy $E(t)$ and estimate the time(s) that **energy** is maximized and minimized.
(Hint use the first derivative test)



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10) Find the exact value of $\int_{-3}^{3} \sqrt{9-x^2}\,dx$ using geometry.

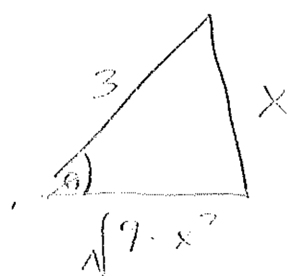
$$\frac{\pi r^2}{2} = \frac{\pi 9}{2} = \frac{9\pi}{2}$$

Find the exact value of $\int_{-3}^{3} \sqrt{9-x^2}\,dx$ using integration techniques.

$\theta = \operatorname{Arcsin}\left(\frac{x}{3}\right)$ $\sin\theta = \frac{x}{3}$ $\csc\theta = \frac{3}{x}$

$\operatorname{Arcsin}\left(\frac{3}{3}\right)$ $\cos\theta = \frac{\sqrt{\quad}}{3}$ $\sec\theta = \frac{3}{\sqrt{\quad}}$

$= \operatorname{Arcsin} 1$ $\tan\theta = \frac{x}{\sqrt{\quad}}$ $\cot\theta = \frac{\sqrt{\quad}}{x}$



$3\sin\theta = x$

$3\cos\theta\, d\theta = dx$

$$\int_{-3}^{3} \sqrt{9-x^2} = \int 3\cos\theta\, dx = \int 3\cos\theta \cdot 3\cos\theta\, d\theta$$

$$9\int \cos^2\theta\, d\theta$$

$$\frac{9}{2}\int 1 + \cos 2\theta = \frac{9}{2}\Big[\theta + \frac{1}{2}\sin 2\theta\Big]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$\frac{9}{2}\left(\frac{\pi}{2} + \frac{1}{2}\sin\left(\frac{2\pi}{2}\right)\right) - \frac{9}{2}\left(-\frac{\pi}{2} + \frac{1}{2}\sin \pi\right)$$

$$\boxed{\frac{9\pi}{4} + \frac{9\pi}{4} = \frac{9\pi}{2}}$$